

REPORT ON PROCEEDINGS BEFORE

PUBLIC ACCOUNTABILITY AND WORKS COMMITTEE

INQUIRY INTO DATA CENTRES

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At Jubilee Room, Parliament House, Sydney, on Friday 1 May 2026

The Committee met at 10:00.

PRESENT

Ms Abigail Boyd (Chair)
The Hon. Mark Buttigieg
The Hon. Scott Farlow (Deputy Chair)
The Hon. Jacqui Munro
The Hon. Peter Primrose
The Hon. Emily Suvaal

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The CHAIR: Welcome to the first hearing of the Committee's inquiry into data centres. I acknowledge the Gadigal people of the Eora nation, the traditional custodians of the lands on which we are meeting today. I pay my respects to Elders past and present, and celebrate the diversity of Aboriginal peoples and their ongoing cultures and connections to the lands and waters of New South Wales. I also acknowledge and pay my respects to any Aboriginal and Torres Strait Islander people joining us today or online. My name is Abigail Boyd, and I am the Chair of this Committee.

I ask everyone in the room to please turn their mobile phones to silent. Parliamentary privilege applies to witnesses in relation to the evidence they give today. However, it does not apply to what witnesses say outside of the hearing. I urge witnesses to be careful about making comments to the media or to others after completing their evidence. In addition, the Legislative Council has adopted rules to provide procedural fairness for inquiry participants. I encourage Committee members and witnesses to be mindful of these procedures.

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Ms BELINDA DENNETT, Chief Executive Officer, Data Centres Australia, affirmed and examined

The CHAIR: I now welcome our first witness. Thank you very much for making the time to give evidence. Would you like to make a short opening statement?

BELINDA DENNETT: Thank you, Chair and Committee, for the opportunity to appear today. My name is Belinda Dennett, and I am the chief executive officer of Data Centres Australia, the peak body for the data centre sector. Our members represent 86 per cent of operational data centre capacity across the country. I wish to make a short opening statement that touches on five points. Firstly, data centres underpin a modern economy. They run everything from banking, transport systems, air traffic control, emergency management systems, visa processing, hospital systems, business applications and the like. They are critical infrastructure.

Secondly, the growth in data centre capacity is being driven by three things: the ongoing shift from on-premise compute to the cloud, which brings with it a 67 per cent energy efficiency; the continued uptake of new digital services; and the rapid adoption by consumers and businesses of artificial intelligence. If Australia's compute processing requirements to 2030 were met by hosting servers in the basements of buildings, or what we call "on-premises" instead of data centres, electricity consumption would be 22 terawatt hours more. Data centres are an efficient way to run a digital economy.

Thirdly, the situation in Australia is very different to the US. The scale is significantly smaller, and the regulatory environment is vastly different. Data centre operators pay their way. The national electricity rules require all large load users, including data centres, to pay 100 per cent connection costs and augmentations to the network. Data centre operators will pay \$10.3 billion in energy infrastructure to 2030, and \$1.1 billion of that is excess capacity available for public use. Data centre operators do not receive subsidies or tax incentives. Fourth, there is significant misinformation around water use in the public debate. The ABS water accounts show that data centre water use is 0.04 per cent of Australia's water use, and less than 1 per cent of Sydney's water use. The 25 per cent of Sydney's water by 2035 that has been widely repeated in the media and in submissions to this inquiry is not based in fact. It was dismissed by IPART—the independent pricing regulator—as being unverifiable, and it should be dismissed.

My fifth point is that the opportunity to attract AI infrastructure investment to Australia is not just an opportunity for foreign investment. The countries that build the infrastructure that supports the greatest technological shift we have seen will capture the jobs, the economic value, national competitiveness and the ability to influence how AI reflects their own values, culture and laws. Those that don't will become a consumer of someone else's technology under someone else's rules. Data centres are not a problem to be managed. They are essential infrastructure, generating the highest gross value added per unit of energy of any industrial sector. We welcome this inquiry and I'm happy to take your questions.

The CHAIR: Thank you very much, Ms Dennett, and thank you to Data Centres Australia for making their submission. Effectively, it seems that you are an advocacy body for the data centre industry across Australia.

BELINDA DENNETT: That's right.

The CHAIR: How did that come about? How long has Data Centres Australia been a thing, and who are your constituents or your members?

BELINDA DENNETT: We launched on 28 November last year, so we're five months and two days old. We came out of five data centre operators who came together around 2½ years ago to talk about how we could get some more interest in the sector, which is a little ironic now, because the interest is obviously very high. A lot of the rules that govern infrastructure were written long before data centres existed. Some of those are not fit for purpose. Those five companies came together, really, to educate and advocate for settings that would help us ensure that we could build the infrastructure here in Australia more quickly. Those five companies worked together in an informal alliance for two years—

The CHAIR: Sorry, so we're talking about AirTrunk, AWS, CDC, Microsoft and NEXTDC?

BELINDA DENNETT: That's correct, yes—three Australian pure-play co-location operators and two what we call "hyperscalers", so they build their own data centres.

The CHAIR: In terms of your entity, Data Centres Australia, other than representing the interests of those data centres, does it have, for instance, a policy function? When you're coming to us here today, is there any independent or separate backing of knowledge or experience, or is it simply "Here's what these companies are saying, and I'm taking all of that and putting it forward"?

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BELINDA DENNETT: We are a membership-driven organisation. All our funding comes from members. We have two tiers of membership: our principal members, who must be operating a data centre in Australia or have firm, committed plans to operate a data centre in Australia; and then we have affiliate members, who are anyone who might be part of that supply chain. That is the Schneiders and Siemens, who make parts of the componentry that go into the supply chain of data centres; energy distributors, who are very focused on how they connect in with data centres. But our core focus is on those principal members in terms of policy advocacy. We say we have two driving agendas. One is that Australia doesn't miss this opportunity—so we say we are advocates for Australia's interests here—and then advocating to be able to build and operate data centres in Australia.

The CHAIR: That second tier of membership, you say it includes those suppliers into the build. Does it also include consulting firms or people who are on the outside of that, as well?

BELINDA DENNETT: Yes.

The CHAIR: Which consulting firms does it include?

BELINDA DENNETT: Mandala Partners, because they've done a lot of work for us in helping us develop research. They are an in-kind partner, so they don't pay to be a member, but they support work for us.

The CHAIR: Who else?

BELINDA DENNETT: DC Byte, who we just did a collaboration with on producing forecasting data. Is there someone you're particularly—

The CHAIR: Well, I'm just curious as to whether EY is part of that, or PwC, or any of those.

BELINDA DENNETT: No.

The CHAIR: Again, I'm really grateful for your submission. The Government didn't make a submission. But also there was a bunch of data centre operators that were going to make a submission but then withdrew, and also witnesses that we invited here today from various data centre companies that withdrew on the basis that you were coming and that you had made a submission, and that that was all that they had to say. In your submission, you talk about how all of these data centres are very different. Do you think it's sufficient that this inquiry only hear from yourself in relation to that entire industry sector?

BELINDA DENNETT: I think that's a decision for operators. We certainly didn't mandate that. We were happy to put in a submission on behalf of our membership. That's what peak bodies tend to do. We don't stop operators or members from having their own say.

The CHAIR: So there wasn't an agreement within that they were all going to not participate in this inquiry, because—

BELINDA DENNETT: No. I actually wasn't aware if they were asked to—whether they were. I have not had—

The CHAIR: Can I get your assurance, then, that Data Centres Australia will be encouraging its members to participate in this inquiry?

BELINDA DENNETT: Well, we would leave that to their own—

The CHAIR: But you are them.

BELINDA DENNETT: Members make their own decisions. But we're pretty new, and it's the first time, I guess, many companies have had a peak body to represent them.

The CHAIR: Previously, you were the lobbyist for AirTrunk, correct?

BELINDA DENNETT: I was the head of government relations.

The CHAIR: We would call that a lobbyist. I guess it looks to me now that, instead of having all of these data centre operators turning up to answer our very specific questions about their operations, we have a lobbyist for the sector appearing instead. What would you say to that? How is your organisation anything other than a lobbyist group?

BELINDA DENNETT: We say we're here to educate and advocate. We don't come and ask for incentives, for preferential treatment. Most of what I do is myth-busting and laying out the facts about the industry.

The CHAIR: On that, one of the things that's often said about the industry is that it is an incredibly secretive one with very cosy relationships with government and with other industry sectors—that it's very hard

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for the public to get any real understanding about. Your appearance here today, in the context of all of the industry players not wanting to participate—how does that bust that so-called myth? What do you say to that?

BELINDA DENNETT: That's an interesting framing that I haven't heard before. Certainly the industry is highly competitive. When you think about data centres competing for customers, for land, for energy, for water connections, it does tend to be quite—I wouldn't use the word "secretive", but certainly highly competitive. That means there's a real focus on not sharing certain information. I think we can do a better job of that, and certainly that's what Data Centres Australia has been trying to do, putting out that forecasting data this week.

The CHAIR: Let's talk, then, about your submission. There's a bunch of parts of your submission and also in your opening statement that are contested by pretty much everyone who is not industry, so I'd like to talk about that. One of the arguments that gets made is that there's this phantom demand, which seems to be quite convenient because, on the hand, you're talking about data centres being this enormous economic opportunity that will bring—and then you go to the best possible case of all of these data centres are going to get built and this is the huge amount of money that is apparently going to be pumped into the economy. But on the other hand, when we're looking at the energy and the water use, we're supposed to assume only some of them get built. Which is it, and why is the industry so intent on trying to downplay the water and energy burden?

BELINDA DENNETT: I don't think we're trying to downplay anything. My role here is to put the facts on the table. The data that we released—and I did send it to the Committee; we only put it out this week. The *Australian Data Centre Forecast Report* is independent data that is trying to show what is real, compared with this concept of phantom demand.

The CHAIR: Who was the report writer for that? Who did that?

BELINDA DENNETT: It was Data Centres Australia in conjunction with data from DC Byte, which is an independent global analyst on the data centre market. What they do is map data centre development across all countries—8,300 data centres they're mapping. They have a number of data points. They know the industry really well. They know the difference between a press release announcement and what is actually getting built. They have on-the-ground analysts that go out to sites and see, "Is this site actually being built? How much of it has been built? What's the phased build-out of that?" which enables them to have far more granular data than we're finding has been put out by property developer companies, by investment bankers, by desk research that looks at press releases and announcements, and grounds that in reality. Some of the things—

The CHAIR: Just to clarify, sorry, because your submission was very much based on two industry-commissioned reports, which were commissioned by the data centres. There was no citing of independent academic research. This particular data that you've now—this is also industry-commissioned data?

BELINDA DENNETT: DC Byte is an independent global company.

The CHAIR: Is it commissioned by industry?

BELINDA DENNETT: It was done as a partnership. We didn't commission it. It's data they have available that anyone can subscribe to and purchase.

The CHAIR: Can we look, then, at AEMO data. You did cite this, but it showed that existing data centres consumed 3.9 terawatt-hours in the '25 financial year but have capacity to consume 13 at maturity, which would be a more than threefold increase from the current facilities alone with some ramping up through to financial year '41. That's before any new approvals. Do you dispute that that's the impact on our energy network?

BELINDA DENNETT: I think if you talk to AEMO—and AEMO were briefed by the analysts and found this data extremely helpful. We can just take a step back—

The CHAIR: Do you have any evidence of that response?

BELINDA DENNETT: —and talk about—

The CHAIR: Are you saying AEMO found it helpful?

BELINDA DENNETT: That was their feedback directly after we'd briefed them.

The CHAIR: To you?

BELINDA DENNETT: Yes.

The CHAIR: But there's no document to evidence that?

BELINDA DENNETT: No.

The CHAIR: No. Okay. Sorry.

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BELINDA DENNETT: But I did just swear an affirmation.

The CHAIR: That's fine. I'm just—

The Hon. MARK BUTTIGIEG: Are we having AEMO, Chair?

The CHAIR: We will. It will be very helpful.

BELINDA DENNETT: I believe they put out a new forecast in May.

The CHAIR: Yes.

BELINDA DENNETT: What AEMO, what the distributors, what companies—sorry, what governments and what utilities have been telling us is that forecasting data is really difficult. What has traditionally been used to forecast energy use—and this was the situation with that Sydney Water figure—is they use applications: applications for energy connection, applications for water connection, applications for approval. Anyone can put in an application for an energy connection. I can assure you, since I started this role, every property developer with a piece of land rings me to tell me they've got a great deal and this piece of land has energy available. So it's a speculative process. Phantom demand is trying to sift through what is real and what is not real.

The CHAIR: On the water, though, because that one in particular—obviously, Sydney Water needs to work out what the absolute capacity would need to be should all those applications be approved. That makes perfect sense to me. If you're a utility, you need to make sure that you've got sufficient water and you—

BELINDA DENNETT: If all those applications were to proceed.

The CHAIR: That's right.

BELINDA DENNETT: But they're never going to proceed.

The CHAIR: How does their figure of 25 per cent by 2035—because your figure is 1.9 per cent.

BELINDA DENNETT: Yes.

The CHAIR: How is it so different? You're saying that, what, a twenty-fifth of these things would be approved?

BELINDA DENNETT: I think you have to ask Sydney Water how they've calculated that 25 per cent figure because there is no methodology put to that whatsoever.

The CHAIR: We will ask Sydney Water when they come in.

BELINDA DENNETT: What we imagine they have done is used every application in the system. AEMO's own modelling suggests six out of seven megawatts of request for connection never get built.

The CHAIR: Where's your methodology for the water claim?

BELINDA DENNETT: It's from the ABS water accounts—metered water use, backed up by CSIRO.

The CHAIR: Sorry, but your forecast? Where is your 1.9 per cent?

BELINDA DENNETT: That's from the Mandala report, modelled on the ABS water figures.

The CHAIR: The industry-commissioned report.

BELINDA DENNETT: Independent economic consultants, yes, paid for by industry.

The CHAIR: They're not independent if they're also members of DCA.

BELINDA DENNETT: Yes, but they have to put their name to a credible methodology, which they have outlined.

The Hon. SCOTT FARLOW: By way of clarification, in terms of your members, I take it that they have been members of other industry groups as well, like perhaps the Australian Computer Society or the Property Council. Would that be the case?

BELINDA DENNETT: Some members have been and continue to be members of a range of different industry associations, yes.

The Hon. SCOTT FARLOW: But those industry associations didn't necessarily advocate purely on data centres. Is that correct?

BELINDA DENNETT: Correct.

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The Hon. SCOTT FARLOW: So not necessarily something which is bizarre in terms of the establishment of a specific data centre advocacy group.

BELINDA DENNETT: I don't think so. When you think of the interest in this sector, when you think of the economic round table in Federal Parliament last year, it was the hottest issue of discussion and there wasn't one data centre company around that table. That was the feeling—that we needed a peak body to represent the interests of the operators who are on the ground. If you look at the submissions to this inquiry, there's a lot of people giving their advice on what data centres should do that is not grounded in what is real about how the industry operates. That's what we're here for.

The Hon. SCOTT FARLOW: With that being said, you outlined that your members, and in fact the whole industry, are competing for land, water and energy. When we're looking at scarce resources for all of those factors, why is it in our best interest in New South Wales to have data centres occupying that land and using that water and energy? What's the economic benefit for the State?

BELINDA DENNETT: Data centres and the associated tech industry that they underpin produce \$12.6 billion per terawatt hour of energy use in gross value added. That's higher than mining; it's higher than manufacturing; it's higher than any other sector. So from a pure economic perspective, it has a very high return of interest. The other thing to think about data centres—

The Hon. MARK BUTTIGIEG: Can you just outline that economic benefit, because I'm not quite sure how increased use of energy translates into economic activity. Can you just articulate in terms of employment, for example, something tangible?

BELINDA DENNETT: Data centres underpin every industry. If you go to a factory these days, you'll find there aren't many people working in them. They are a lot of robotics and automated machines. Those machines are run through data centres. Data centres automate all of that.

The CHAIR: Some of the data centres, though. Not all data centres are for that.

The Hon. MARK BUTTIGIEG: This is not a rhetorical question; I just genuinely want to know. The position is that it gives New South Wales—and, more broadly, Australia—a comparative advantage if we have domiciled data centres here. I genuinely don't understand this. Is there an offshore market where, when we use Copilot with Microsoft, are they all resident servers or is some of it overseas?

BELINDA DENNETT: It will depend on the service. When I started working at Microsoft 14 years ago, the view was there would only ever be data centres in North America and that's where services would run from. A couple of things have changed. Firstly, governments actually want data hosted onshore. We see a lot of that, whether it's legislated or not. There is a preference for that.

The Hon. MARK BUTTIGIEG: And why is that, Ms Dennett? What is the big comparative advantage of having it domestic? Presumably you have stronger signal strength and broader bandwidth.

BELINDA DENNETT: Latency is very important and that was my second point. The way we're using digital services now—and increasingly with AI, where you're back and forward—you need really low latency. Think about some of the things that are done out of data centres. Air traffic control needs super low latency so there is no delay. Having data centres onshore reduces that latency, but it also gives you a whole lot of things. When you think about health data, there is a preference from governments and from consumers that they know that data is held onshore. Some of that is just perception that it's somehow safer, but I think there's also a view that we want it under Australian laws, not hosted in another country where the laws are different. And that's one of the reasons why Australia is such an attractive market for data centre investment, because we're a strong, politically stable, law-abiding country where people feel safe about data and applications being hosted.

The Hon. MARK BUTTIGIEG: I'm a bit dubious about this thesis, because you think about the world financial system and our banking—we're doing this stuff across borders all the time, every microsecond. So I don't really get that whole sort of security—

BELINDA DENNETT: There's data in transit and the whole point of digital economy is that data can flow freely. Data is encrypted in transit and then you have data at rest. A lot of people—and this is why I say it's not necessarily factually based, but it is how people feel. People feel their data is safe if they think it's on their shore.

The Hon. MARK BUTTIGIEG: So the nub of the economic advantage is the latency and the speed with which you can process by having these things domiciled here. That's really the economic benefit.

BELINDA DENNETT: No, I think there are many others in terms of the value. A data centre is over a billion dollars in cost to build, so there is a lot of economic value in building data centres. There's pull-through of

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the value chain, the suppliers, the concrete manufacturers—those sorts of things that we build an industry around in building data centres here—and then AI changes the whole system again. People really are talking about, you know, AI is changing how you build a data centre. It's changing how a data centre operates.

The Hon. MARK BUTTIGIEG: There's obviously the lead-in construction economic uplift and activity. What about ongoing employment and maintenance? It's at a pretty low level, is it?

BELINDA DENNETT: Data centres run 24/7, 365 days a year, so there are operational staff working in data centres every single day. You have around, per megawatt, one full-time employee per megawatt. You now have data centres at 100, 200, 300 megawatts, so that's the sort of employment numbers you're talking about. But we also have some of the world's best data centre companies built and headquartered here in Australia that employ all the business services that support companies when they're headquartered here as well. We also think that, in terms of building AI, we're starting to see what I say about infrastructure: You lower the barrier to entry. We're starting to see Australian, homegrown AI companies, AI factories, firms of the world. We're seeing Sharon AI, companies like that, that are building a new part of the tech stack, if you like, based here in Australia.

The CHAIR: Does DCA support mandating local content requirements?

BELINDA DENNETT: I think that gets difficult. We would need to work with the construction teams. What we would say about construction is that certainly there's lower cost in building locally. Part of having more of the industry here is you do create those flow-on companies that have more business here. We need to separate the building of the shell and core, as we call it, and the compute. The compute is very much imported and that's owned by the customers. The hyperscalers, the customers, bring that equipment in. That's not a data centre operator.

The CHAIR: I understand that, but how then does that benefit the local economy? If we're bringing everything in from—

BELINDA DENNETT: Not everything, but the compute.

The CHAIR: But if we bring a lot of things in from overseas—again just on this economic benefit point, because in my view and in the view of most of the other submissions it's quite inflated—how do you calculate what you think is the local benefit or the benefit to Australia in any real terms?

BELINDA DENNETT: I think there are a number of parts, and we haven't talked about the part of Australia being an exporter of intelligence. When we get to AI, it produces intelligence.

The CHAIR: Again, in terms of the actual figures, though, there's a lot of figures being bandied about by the industry.

BELINDA DENNETT: Yes, and they are calculating a number of different parts of this, from the construction and building of a data centre to the operation of a data centre, and then the opportunity to export intelligence and tokens.

The Hon. SCOTT FARLOW: Just to that point and the sovereignty argument, effectively, why is it important for Australia to have a data centre industry, and why is it important for New South Wales, in particular, to have one?

BELINDA DENNETT: If you don't look at what you lose, there's a conflation of "If you stop data centres, you'll stop AI." We get that. There is a real concern in the community about artificial intelligence and what that means for society. But we won't stop AI; that is coming. So we either become an importer of someone else's technology that has no Australian culture, values or laws built into that, or we build that here and we have some say and control over what that looks like.

The Hon. SCOTT FARLOW: What is that say and control? What does the Australian values AI look like? What can we do to govern how the systems operate?

BELINDA DENNETT: I think if you are operating here, and if you think about some of the companies who have talked about setting up a major base here in Australia, employing Australian researchers that then model on Australian conditions and Australian language, using Australian language to make sure AI thinks and speaks and reflects our stories, our laws—if you're operating here, you're governed by Australian laws—

The CHAIR: Sorry, this is one of those points around sovereignty that I'd love you to clarify. Physically, the data centre is in Australia, but it is OpenAI, Microsoft, Amazon and Google who are actually using the data centres. None of those are a sovereign Australian company.

BELINDA DENNETT: But they're still governed by Australian law. They're still required to meet—if you are based here and operating here, there are a whole lot of requirements on you as a company.

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The CHAIR: So you think we can influence their use of technology, AI and their use of data, simply because they—

BELINDA DENNETT: Operate here. Absolutely.

The CHAIR: —operate here, because they happen to use, for part of their global data storage, a location in Australia.

BELINDA DENNETT: It's more than storage. I think understanding—

The CHAIR: And processing.

BELINDA DENNETT: Processing and building models and employing local people and having the headquarters here and being bound by Australian laws.

The CHAIR: There's no data borders where we say, "Well that can't be accessed"—

BELINDA DENNETT: No, but if you're going to be doing that, you tend to come and have a base here. Microsoft has been in Australia for 40-plus years. It has people here. It employs people. It has a headquarters.

The CHAIR: Sure, but it's not Australian, from a sovereign—

BELINDA DENNETT: No, but it very much is bound by Australian laws. It is required to be bound by Australian laws if it's operating here.

The CHAIR: But not in relation to its entire global operations.

The Hon. MARK BUTTIGIEG: It's a good question. When I'm typing into Copilot, "I want you to analyse this document," my understanding is the document goes onto their server and they have custody over it. They say they delete it after a while, but we don't really know. That's a function of what the Chair is talking about, isn't it? Because then, if it was subject to domestic laws, we might say you have to make that the custody of the user and delete it after 10 seconds, or whatever. That's a practical example of what I think the Chair is talking about.

BELINDA DENNETT: I think you would need to talk to Microsoft about exactly what happens in those operations.

The Hon. MARK BUTTIGIEG: Exactly. That's our point, right?

BELINDA DENNETT: My point is also that's not something—a data centre operator doesn't touch the data or the compute.

The CHAIR: That's the point, though, isn't it? Firstly, there's no transparency over who actually are the customers and users of these centres a lot of the time. We know who the proponent is, who the operator of the data centre is. But when it comes to this idea that somehow then we have data that is in Australia only, that's just not true, is it?

BELINDA DENNETT: Well, we were just talking about data in transit. The whole premise of a digital economy is that data can move.

The CHAIR: Exactly. It's part of global capacity for data centres. So how, again—

BELINDA DENNETT: But you still have data processed and stored and transmitted. Data centres are heavily regulated by Australian legislation. The Security of Critical Infrastructure Act, the Hosting Certification Framework—these are things that have real bearings on operators and also on the people that own the compute—the processors.

The CHAIR: We can talk about how strong the regulations are but, just coming back to Mr Buttigieg's original questioning, part of our terms of reference is that there are different types of data centres that are doing different things and there are different uses for them. I think no-one on this Committee doesn't think we should have some data centres, and we need them for certain things. The question is how many do we need and what is actually beneficial, on balance, for the State and for Australia. We can accept that we want to have, for example, Australia's security information securely stored somewhere. But when we're talking about the broader ecosystem for data centres, where you've got AI and other applications, sell to us why we should be trying to get as much of that business as possible in Australia. What's the economic benefit to us of that?

BELINDA DENNETT: As I've been trying to explain, AI is the big driver of growth in capacity. At the moment, if you look at the rate of both AI uptake in Australia—Australia is the fourth largest global user of Anthropic's Claude, an enterprise-grade AI, so Australian businesses are adopting AI in record numbers.

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The CHAIR: We could have no data centres and still have access to that, so what's the benefit of the data centres?

BELINDA DENNETT: Well, it's going—

The Hon. EMILY SUVAAL: Just hypothetically, to flip that around another way, what would happen if we had no data centres here?

BELINDA DENNETT: We would just import that from overseas, probably from Malaysia, probably from Singapore—somewhere close enough that there isn't incredibly poor latency.

The Hon. EMILY SUVAAL: So the concept that we could at least not introduce some new data centres in New South Wales is perhaps a fallacy. What I'm trying to get at is, there seems to be a level of resistance to the concept of having data centres here in New South Wales but, conversely, we can't not have them. Is that right?

BELINDA DENNETT: If I can cast your mind back a couple of years ago when there was the CrowdStrike outage, that was not a data centre, but it was a piece of the tech stack. That crippled the economy. It was on a Friday afternoon here. It was midnight wherever that incident happened. It took out people's travel plans. People couldn't access their money. The economy runs on digital, and that is underpinned by data centres operating 24/7.

The Hon. MARK BUTTIGIEG: Just from a practical point of view, with that Malaysian example, what's the comms infrastructure? Is it submarine cables or wireless or both?

BELINDA DENNETT: Subsea cables.

The CHAIR: But, again, how many do we need? To Ms Suvaal's point, maybe we need one, maybe we need two.

BELINDA DENNETT: We talk about data centres in megawatt capacity as opposed to number of data centres. Ten years ago data centres were one to five megawatts, and we've got a lot of very small data centres here. What we're now starting to see is much larger data centres, so capacity is probably a better way to talk about it than talking about numbers of data centres.

The Hon. MARK BUTTIGIEG: Back to my question, that means you're actually physically limited by submarine cables, in effect.

BELINDA DENNETT: You could be. There are submarine cables being laid all the time. Google, SUBCO—there are a number of Australian companies and global companies building connections both to Asia and to the US, so there are increasing numbers of submarine cables. You need landing stations, so you need a small data centre at the point where they come in, to make sure you're processing that data, and then that's fed into the capital cities.

The Hon. MARK BUTTIGIEG: Presumably it's cheaper to build data centres than lay massive submarine cables every 10 minutes, or whatever?

BELINDA DENNETT: Yes.

The Hon. SCOTT FARLOW: This is not some isolated phenomenon that's happening in Australia. This is happening all around the world. To that discussion about what we've seen, we have seen different iterations in terms of the data centre development—we've seen cloud computing, we've seen more onshoring as you were talking about in terms of regulations and Australian governments wanting to onshore data. Now, we're getting to this global proliferation in terms of data centres, because of the processing needs of AI. To that point, what are the advantages for Australia and for New South Wales in being able to harness and attract data centre development here instead of offshoring into Malaysia, having Asia feed in to Australia?

BELINDA DENNETT: You're absolutely right. It's a global race and countries all around the world are trying to attract this investment. The big change was the US said, "We want to win the AI race," and the way you do that is build infrastructure. There is a lot of competition. If you look at the global companies, they are spending around \$800 billion this year alone in building infrastructure. A lot of that is forecasting the demand that's going to be needed. We're right at the start of the AI transition. It will play a role in every part of the economy. Every industry will be transformed by AI. AI produces new data, it produces new code, it compounds on itself. You have AI producing data.

That is part of the reason why we need more capacity. It's the reason why countries are competing so fiercely to attract this investment. I said in my opening statement, Australia doesn't offer tax incentives. We're not asking for tax incentives. But many countries are offering enormous incentives for people to build in those markets. A number of our Australian companies are building in Malaysia, in India, because there is huge demand

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for that. So New South Wales and Australia are competing globally for this investment. This investment brings \$142 billion per year—think of what that can do in terms of Medicare, hospital services, the like. It's not very often a country gets an opportunity to find an investment source like this.

The CHAIR: Sorry, how does the \$142 million in construction costs allow us to fund Medicare?

BELINDA DENNETT: Investment costs into the country. If you think about how do governments get revenue, it's either through taxpayers paying tax or it's businesses paying tax.

The CHAIR: So \$142 million in tax revenue?

BELINDA DENNETT: Economic value.

The CHAIR: Goldman Sachs has found there is no meaningful relationship between AI and productivity or data centres and productivity at the economy level. Again, perhaps on notice, rather than me taking up more time, you could just give us exactly how you've calculated the \$142 million.

BELINDA DENNETT: Well I didn't, but I can certainly share the report where that's calculated and all the methodology is there. I can break it down for you. It was \$118 billion in productivity value of AI being adopted across every industry. There is an export value there of around \$12 billion, exporting intelligence, exporting tokens. Then there's creating new Australian AI and what that does creating new businesses.

The CHAIR: The tech companies are famously resistant to being taxed. Google threatened a capital strike if it got less favourable tax treatment.

BELINDA DENNETT: I think the tax is quite important to Australian accounts.

The Hon. JACQUI MUNRO: Can I pick up on that please from a State level? Personally, I think we need more data centres and I think they're a fabulous way for us to enliven our economy. Part of the problem has been a lack of coordination between government and data centre operators and the community to explain how data centres are going to be strategically located, how they will utilise public infrastructure, water and energy, and how they will be able to contribute back. In that, you've got planning considerations related to land tax. I was curious about what interaction you've had with the Investment Delivery Authority in relation to data centre applications.

BELINDA DENNETT: We have been working with the agencies that make up the IDA around what this actually means, so how does this IDA process—it was announced to speed up the process of planning applications, which in New South Wales are notoriously slow, two-plus years, so we're still working through how that actually delivers. They had a process. They called for applications and they announced which ones were in the IDA. We're not quite sure now what happens, or we now just go through the same planning process. It's not clear to me how that is sped up. There are lots of conversations happening around these new data centre principles. Again, we probably share your questions about how the IDA is going to deliver on what it was stated to do.

The Hon. JACQUI MUNRO: When you say you're in touch with the different authorities, are you separately speaking to Treasury, separately speaking to Investment NSW and separately speaking to DCCEEW?

BELINDA DENNETT: There have been some meetings where those parties are coming together, which I think is really helpful, along with the utilities, along with operators. We've found that is probably what needs to happen more because, when you talk about how utilities think about planning, those one-on-one conversations aren't getting us to where we need to be. So this idea of more information sharing as we all plan—early days, but starting to show some fruit.

The Hon. JACQUI MUNRO: How were those projects selected that have been given some IDA tick of approval? What is that? Is there any rhyme or reason to why those particular projects have been chosen, to your knowledge?

BELINDA DENNETT: I think you would have to ask them.

The Hon. JACQUI MUNRO: Are there conversations around how land tax might relate to developments either being drawn into particular areas or that there would be land tax treatment over the longer term that would be different to what already exists in legislation?

BELINDA DENNETT: I haven't had any of those conversations. We certainly are not lobbying for any changes. Our position has been that data centres have a number of requirements about where they locate. Government dictating where a data centre builds is not ideal. There's already planning and urban design, and areas where we can apply to build, and we think that's appropriate.

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The Hon. JACQUI MUNRO: On the idea of something like innovation precincts, do you or do your members think that there is a benefit to being in an area that might be designated something like an innovation precinct, or is it not front of mind for operators?

BELINDA DENNETT: I think what could be looked at is that data centres tend to cluster. They cluster around where there is energy availability, land availability. There's also a requirement that data centres are quite close to each other, because they use resiliency and backup if something was to happen. They tend to cluster, and I think there is great opportunity to think about how you build where they already are going into more of a precinct, what else could be around those areas and how could you connect the land between. That hasn't been done and that probably goes to the competitiveness of the industry. They work very much on their own, but perhaps that's a role now with Data Centres Australia. We can look at more of that precinct planning.

The Hon. JACQUI MUNRO: Have you been asked to contribute to or engage in any process related to a data centre strategy?

BELINDA DENNETT: No. We hear a lot of comments about a data centre strategy, whether it's Sydney Water or whether it's someone else, but they have not come to us, no.

The Hon. JACQUI MUNRO: In terms of data centre operators either being required to or proactively generating their own energy, for example, or at least energy to put back into the grid, could you outline if that's happening, what's happening, and the potential future pathways to essentially generate better social licence around energy usage in particular, and generation?

BELINDA DENNETT: It's really important to understand what the sector already does. Data centre operators, in conjunction with their customers, sign power purchase agreements, which are long-term offtake agreements with renewable energy projects. They underwrite those projects. They make those projects viable. We calculate on PPAs, those particular financial instruments, and large generation certificates, that around 70 per cent of data centre energy is offset by renewables, bringing 1.5 terawatt hours of new renewable energy to the grid. AWS just announced two weeks ago nine new projects they're backing, including battery storage, to offset energy use across their portfolio. Most of our operators and their customers have net zero by 2030 targets. Data centres, as I said in my opening, are 67 per cent more energy efficient. Data centres came out of a need to be more energy efficient by sharing resources.

It is a forward-leaning industry. Many of them have sustainability-linked loans that require them to meet certain targets. I think there is a desire to get to 100 per cent, but we need more viable renewable energy projects that currently are also stuck in planning. It's not a lack of desire to do that. Trying to match a renewable energy project timing and a data centre timing is difficult. Then the other thing we would say is that when you think about the energy use of a data centre, around 80 per cent of that energy use is for the compute and the processing; 20 per cent is operating the data centre or the infrastructure load. That's why they're done in conjunction. When people are like, "Data centre operators should go," or, "They only pay for the 20 per cent," that's why they're done together.

The Hon. JACQUI MUNRO: On the specifics of using renewables for backup power, are there ways that the industry is exploring how that can occur? Do you need to have, essentially, a non-renewable source of energy connected to the site for backup purposes, or are there other ways, potentially internationally—I don't know, geothermal or hydro—that you can connect?

BELINDA DENNETT: Data centres are connected to the grid, and then they will purchase renewable energy to come into the grid. Data centres require a very stable energy load, 24/7, which is why firming becomes so important. Battery storage becomes really important. If you talk to the AEMO, the distributors, data centres are very good for stability on the grid because they're that steady offtaker on those days when we have too much solar coming in. Data centres don't rest on public holidays; they soak up the excess, so they're actually very good for the grid. Backup—because a data centre can't go down, if there's a grid interruption, data centres have generators on site as their backup.

They're essentially an insurance policy, just as hospitals have backup generators, just as our banks have backup generators. They're used very sparingly. Battery storage may get to the point where that's the alternative, but at the moment it's not long enough duration. If you had an outage for eight hours, that would be a problem. That's what people are starting to look at. Geothermal, gas backup—they're options that are looked at around the world. I think what we're seeing is so much innovation in this space, and at pace, that things will change and they will change quickly.

The Hon. SCOTT FARLOW: Sitting in your seat a few months ago, the Treasurer was there, and the Treasurer said that he didn't see us in competition with Victoria and that New South Wales was a premier destination when it came to data centres. Just interested in your thoughts between New South Wales and Victoria

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and the arrangements between both States, and whether New South Wales is seen as a premier data centre destination and, if so, why?

BELINDA DENNETT: Let's start with the data. The data tells us that 96 per cent of Australia's data centre capacity is in Victoria and New South Wales. New South Wales has around 60 per cent of that capacity, maybe a little bit more, so New South Wales has been the leader. I think Victoria is becoming far more attractive, partly because of their ability to approve processes quickly—land availability, there is more energy availability in Victoria—but the thing I will keep saying is I love Australia and its State of Origin wars but it's not where the global race is. You are competing with Malaysia, with Singapore, with Japan, with India to attract these big workloads. If we focus too much on Victoria versus New South Wales, we're missing the real game.

The Hon. SCOTT FARLOW: But in terms of the arrangements between the States—and I think you mentioned the planning processes available in Victoria compared to New South Wales—in order to be globally competitive, what needs to be done here to change that equation?

BELINDA DENNETT: We think New South Wales could look at what Victoria is doing in terms of their investment team, facilitation team. They have people who know data centre. Within Invest Victoria there's a team of people that our operators will work with daily that know their business, that know their operations, and they act as a concierge to fast-track. The process is no less rigorous than what it is here in New South Wales, it's just not stuck in people's inboxes, it's not paused for several weeks. They keep the process moving, and that allows our operators to get through that process quickly. Why that matters: A big global company will not sign a contract with a data centre operator unless they've got approval, energy connection and water connection. The longer those things take, the operator is looking at India and—sorry, the customer is looking elsewhere and that's where they'll go because, right now, speed to market. Customers used to want a data centre capacity in two years; they now want it in six months because the demand for AI is so high that they need capacity more quickly. We need our processes to move more quickly.

The Hon. EMILY SUVAAL: Just in terms of where you're looking at locating data centres. I'm just wondering what specifically goes into informing that, and whether there are other areas in New South Wales that perhaps could be—what would encourage you to do so, for example?

BELINDA DENNETT: I think there's a couple of things. Firstly, different data centre types that do different workloads are suited to different locations. At the moment in Australia, the data centres mostly are cloud and they're being built for AI inferencing. What they need is low latency. What they need is proximity to their users, proximity to a skilled workforce, energy connection, water connection. The customers—so those anchor tenants that provide the underwriting to make a data centre project viable—tend to dictate where they want to be. I talked before about clustering. Often they want to be close to where their backup resiliency is so they'll draw a circle on a map and say, "We need you to find us somewhere within this circle." That's why we see the western suburbs of Sydney and the western suburbs of Melbourne are really where the global companies have built their networking infrastructure. That's where they want to be.

Now, if we were to attract some of those global AI training facilities that we talked about earlier that can have less reliance on latency, they could go further out. Certainly, we're seeing Perth as a destination where now data centres are starting to build, or there's a specialised workload so we're supporting mining, you might get a smaller data centre to support that. Different workloads will go to different places and have different needs. But at the moment, where the focus is, is Western Sydney and western Melbourne because that's where the availability zones of the cloud providers is, where they've built their networking and where the natural attributes that we've talked about have been identified.

The Hon. MARK BUTTIGIEG: We're almost out of time, I know. With the actual hardware, the chips and all that sort of stuff that's coming in from Nvidia and all these American companies, I understand the future for us is digging it up and selling it to them and then bringing it back, right? That's kind of where we're at at the moment. But is there a view about our capacity to do this all domestically and get into that market?

BELINDA DENNETT: I think it would be difficult, not impossible. We obviously have the critical minerals. Whether we could create a chip fabrication plant, I don't know. That's probably outside my specialisation.

The Hon. MARK BUTTIGIEG: Why is that? We've got a well-educated workforce; we're very capable at everything we turn our hand to. Why isn't this a growing market?

BELINDA DENNETT: Some years ago we stopped doing manufacturing here, and that's essentially what chip fabrication would be.

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The Hon. MARK BUTTIGIEG: Is chip fabrication a function of your ability to systematise an automation with robotics and things like that, or is it still physical labour?

BELINDA DENNETT: I'm not sure. That's probably outside my expertise.

The Hon. JACQUI MUNRO: It's a scale issue.

The CHAIR: That does bring us to the end of this session. It was unfortunate that you had to go first, because I think a lot of the detail and information around what are data centres and things like that would have been useful beforehand, but that's on us. Apologies for that. We are very grateful for your attendance. If there are any supplementary questions or questions taken on notice, the Committee secretariat will be in touch in relation to that.

BELINDA DENNETT: Thank you for your interest.

(The witness withdrew.)

(Luncheon adjournment)

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Ms CLAIRE PULLEN, Group Chief Executive Officer, Australian Writers' Guild and Australian Writers' Guild Authorship Collecting Society, affirmed and examined

The CHAIR: I now welcome our next witness. Happy May Day.

CLAIRE PULLEN: Thank you, comrade.

The CHAIR: Would you like to start with a short opening statement?

CLAIRE PULLEN: Warami, everyone. Thank you, Committee members, for this opportunity to give evidence today. I note at the beginning that I'm a member of the Writing Australia Council and also an adviser for one of the interested consortia applying to be the successful applicant to run the second New South Wales film studio. So, for the Committee's clarity, I make my remarks today in my capacity as the CEO of the Writers' Guild and our associated group. The Australian Writers' Guild is the professional and industrial body for Australia's performance writers working in screen, theatre, comedy, podcasting, audio drama and game narrative design. We have members in every State and Territory in Australia, but our largest membership group is based in New South Wales.

The Australian Writers' Guild Authorship Collecting Society—one of our group—was founded 30 years ago this year by screenwriters to collect and distribute screen royalties to screen authors in Australia and Aotearoa, New Zealand. We have collected over \$35 million, because every time someone watches anything Australian anywhere in the world, if it's *Bluey*, if it's *Homicide*, if it's *Happy Feet* or if it's *The Lord of the Rings*, screen authors are paid a royalty that we collect and distribute. The creative economy that we're very proud to be part of is growing, and contributes \$67 billion a year to Australia's economy. Creative employment is growing at a rate more than 50 per cent higher than the rest of the workforce, despite some sectors being hit very hard by COVID. New South Wales benefits from a great deal of this activity.

The use of generative AI, particularly the use of large language models like ChatGPT or Claude, put all this at risk. As litigation overseas has demonstrated, these LLMs are built on stolen works. Creators whose books, plays, screenplays, songs, music, TV and film works make up the foundation of these models had those works taken without consent, compensation or credit. These models do not work without the copyrighted works of my members and other creators. We are not aware that any creator or rights-holder group in Australia has been asked for a licence by an AI or tech company to use these works, licensing being a very standard form of operation in our industry—indeed, what we do every single day.

AI data centres will be used to host these models and this data made from my members' stolen work, and citizens of New South Wales will be using these models, likely completely unaware that they are being asked to expose themselves to the risks of copyright infringement by doing so. These models then produce outputs that compete directly with my members for their jobs, and create alternative, royalty-free products. The New South Wales cultural policy, *Creative Communities*, outlined a positive vision for our creative and night-time economies. The New South Wales Government has been a leader on regulating the harms of AI, and can be here, too. As the Committee is no doubt aware, competition for land use, particularly in Sydney, is intense. Creative workers are no different to anyone else facing a cost-of-living and housing crisis, and are increasingly priced out of any of the metropolitan areas near where they work. Expanding housing areas are already under-served by things that need land, like schools or emergency services and health services. Data centres will simply make this problem worse.

New South Wales is also in dire need of a second film studio and, noting my earlier declaration, wherever this film studio is located, this is a preferable use of land, from our point of view, over a data centre. A film studio generates tangible economic benefits, unlike the mostly speculative benefits of artificial intelligence. I end like a good arts practitioner with a provocation. Please put the proponents of these data centres to as much scrutiny and as much microscopic examination as my members are put to when they are seeking a \$5,000 grant to write their screenplay or translate their play into another language or apply for studio time to make something that will be shown all over the world, returning both economic and cultural benefits to New South Wales.

The CHAIR: We had the Data Centres Australia witness here this morning talking about the supposed benefits of data centres. I think we can all agree that some data centres are useful and necessary and we could perhaps get behind that. But there is a lot of confusion and lack of clarity as to what each of these data centres is for. In order to then determine what the overall economic benefit or societal benefit would be, it is hard for us to work that out. In that context, one of the comments that was made was that there's a need to have sovereignty over data in Australia and this idea that, if we have data being housed here, we can regulate the way in which that data is treated. What would be your reflections on the ability of government, whether it's Federal or State, to adequately regulate what happens to our data with those global companies?

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CLAIRE PULLEN: I have a couple of thoughts on that. Obviously, there are physical capacity constraints around the transfer of data through things like submarine cables, so there is a surface argument to be made that housing data centres here is more useful because we'll be physically closer to the data. One of the things I have observed that we don't hear a lot from the big tech proponents of data centres and AI more generally is a meaningful commitment to the concept of sovereignty, if I can be a bit provocative about that. They're a bit half pregnant about it. This is not government building data centres. It's not owned by Australians or New South Wales taxpayers because the government is not building or owning any of these. It's only sovereign to the extent that it exists within our territorial boundaries. That's the first part. If this was truly a benevolent position from the big tech companies, these would be built and then donated to the people of New South Wales or Australia, frankly.

There's an argument, I think, that AI is a general-purpose technology, like electricity or the internet. Our governments used to build power stations and coalmines. I'm not going to get into whether or not the sale of those assets is a useful thing to do, further down the life of those, but I would say that I think it's worthwhile being sceptical about the concept of sovereignty when it's a foreign big tech company saying this is sovereign in some way when there's no meaningful ownership. Certainly I don't understand there was any offer this morning to allow an equity stake from the New South Wales Government in any of these companies that will either build the thing or house data here. The second part of that is, just because you could regulate something, I don't think any of these companies are arguing that they would enjoy that or would advocate for that. I spend a bit of time in Canberra, because copyright law is mostly a Federal issue. They are working very hard and spending quite a lot of money to try and weaken Australia's copyright laws. So I think it's, on the surface, correct to say that you could regulate here, but I don't think it's what any of the big tech companies actually want.

The CHAIR: It's proved difficult in practice when it comes to things like, as you say, copyright.

CLAIRE PULLEN: Yes, it has proved difficult. The Federal Government has done what we think is the correct thing, the thing that ensures there is a future for Australia's creative economy, which is affirming that our copyright laws will not be weakened. It's something that other governments around the world are following. As an example, the UK have actually relooked at their views on AI because of Australia. I was in Canada earlier in the year, specifically to talk about the Australian Government's role in affirming the importance of copyright. So I would say, yes, it's disingenuous to suggest, just because it could be regulated here, that tech companies have a neutral position on what the regulation would actually be.

The CHAIR: You mentioned that data centre proponents and advocates are putting out the idea that it's just like another utility; they're saying that we need water, we need energy and, clearly, we need data centres. When it comes to looking—and this is why it keeps coming back to how important this is—at what these data centres are being used for, I think there is a broader acceptance of data centres that are used for data storage versus data centres that are used to replace workers or even to train large language models and those kinds of things. In that way, it's not all just water—some of it is water but some of it is perhaps a bit of Fanta or beer. Maybe some of it is necessary and critical; others aren't. What is your view on the secrecy around how much of it is to go to something other than data storage? What are the broader societal impacts of allowing a proliferation of those sorts of "Fanta" data centres?

CLAIRE PULLEN: It is a symptom, I think, of the broader problem that you've identified: It's so much in the commercial interests of the big tech companies to keep as much of this secret as possible about what the models are doing, how they're doing it, what they're built on and where the information comes from. One of the key asks for our members—and we regularly speak to our members about AI—is that they would like their works removed from the models. That is the point when I think we could have a more meaningful conversation about licensing, because at the moment there are two issues. One is that the works were taken and so there needs to be some restitution in some form for that, and then there's the ongoing use piece. Without any transparency around what they're planning to do with the data that's already been gathered, it's hard to have the second, more fruitful and probably more economically beneficial to everyone conversation about licensing and ongoing payments.

I suspect it's a similar sort of thing that you'll get pushback around—that it's all a seamless cloth. It comes in and it goes out, and there are commercial arrangements in place, all of which goes to the social licence question. Australians are among the most sceptical people about AI in the world, which I think is a good thing. That speaks highly of our national character, frankly. There is a lot that the big tech companies could do to resolve that anxiety, to fix those questions that Australians have. A relatively low-effort piece for government, frankly, is just to say, "Until you have made these disclosures or quarantined these particular models or types of data, your planning approval won't go ahead for the data centre," for example.

One of the things that is in our submission that we're concerned about is, if you cite data or models in Australia that don't comply with Australian copyright law, anyone who uses ChatGPT or Claude is potentially opening themselves up as an individual to action around copyright infringement. I would say most people don't

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know that. I say this as a copyright law nerd: Normal people shouldn't know more about copyright because it's not interesting in a lot of ways, everyone's busy and it's not an area of law that I think a normal person should be interested in. It's a deeply abnormal area of law.

So I think the expectation from some proponents of AI as an entirely beneficial technology is that it puts the responsibility on the consumer or the user at the very end to make a discernment about it, whereas we could go slightly further back in the process and say, "Can you tell me that the data that's going to be stored here in this data centre complies with Australian law?" which would cover things beyond copyright—child sexual abuse material, for example. "Are you hosting a data server for a dark web drug-dealing operation or the illegal arms trade?"

If you're going to make the argument that AI is a general-purpose technology like electricity or the internet, it doesn't necessarily follow that you can't regulate any harms coming from them. The illegal arms trade is facilitated by the internet and electricity. No-one's suggesting that you turn off electricity or the internet, but you do regulate the harm, and I think it's a similar argument here. If it's a general-purpose technology and it will be generally useful, then you can mitigate the harms. The New South Wales Government has done some of that, and I think that approach is a good place to start. Sorry for the very long answer.

The CHAIR: No, it's very good. We're appreciative.

The Hon. MARK BUTTIGIEG: You expressed the view that New South Wales has taken a lead in managing the fallout of the harm from AI. Can you elaborate on that a little bit?

CLAIRE PULLEN: Yes, I'm referring specifically here to the law against image-based abuse.

The Hon. JACQUI MUNRO: That was a Coalition bill.

CLAIRE PULLEN: New South Wales Government—look at us go!

The Hon. JACQUI MUNRO: Well, New South Wales Opposition.

The Hon. MARK BUTTIGIEG: I didn't actually say NSW Labor, but anyway.

CLAIRE PULLEN: I think I did, from memory, but we can check the script on that one. But it is a really good example of how AI allowed for the generation of a particular type of abusive material that was not available before, and you can legislate against the harm. That's a really straightforward example of that in a New South Wales context. At a Federal level, we've worked really closely with Screen Australia, who is the main funding body for a lot of my members and for the industry generally, around artificial intelligence. There's a set of declarations you're required to make when you're applying for funding from Screen Australia about your use of artificial intelligence. It's a really simple weeding-out tool to say taxpayer money won't be funding something that could put a bunch of creators out of work and also might not comply with our law. So there actually are practical things you can do. If you form a list of the harms, you can work backwards to what the solution might be. That's an example of things that you can actually do at a meaningful level.

The Hon. MARK BUTTIGIEG: In terms of the copyright law you outlined, which is largely the purview of the Feds, what other things can we as a State jurisdiction do to fill in the gaps and stop that IP theft that you've articulated?

CLAIRE PULLEN: I think one of the things that came out of an earlier inquiry—which, from memory, was around education but did have terms relating to artificial intelligence—was an undertaking from the New South Wales Government to engage with their Federal counterparts around copyright. That is actually really useful, as New South Wales is the largest economy in Australia. We are a powerhouse of economic activity, some of which goes back to intellectual property. In terms of what New South Wales State Government could do, include terms about the data and the models in the planning requirements for approvals for the data centres—so, "To the best of your knowledge as the builder of a data centre, do you know that what's going to be in there complies with copyright law and other Australian and New South Wales law?"

The example I gave about Screen Australia—New South Wales has creative funding bodies as well in Create NSW. There's a similar, I think, quite straightforward option to say funding should only go to projects where we know or can be reasonably certain that taxpayers aren't funding something that's been built off AI that's built on stolen work. I'm not an expert in planning law. I want to be clear that my nerdery does not extend that far. But in terms of planning, I suspect there would be a way to give preferential treatment around planning and procurement to tech companies that are prepared to certify that all the data in their centres is ethical and that models that they're running in their centres are free of stolen work and that they're able to give some sort of reasonable assurance around that.

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This is an easy one, but use land to build a second film studio. I don't want this to be like our second airport where we just complain about it for 30 years before we actually do it. We are in dire need of more studio space somewhere in New South Wales. As I said, I'm an adviser for the Central Coast bid. I would love to have it located near me, but—

The CHAIR: Hear, hear!

CLAIRE PULLEN: I'm in the right room for that particular position. Genuinely, we need a second big studio and, if it was a choice between a data centre and the film studio, I'd say film studio, please. The other thing I've been talking about a lot with the Federal Government is building more capacity at a government level around how you can share ideas, share insights around the problems that AI is going to present. One of the issues we talk about a lot at the ACTU, for example, is the rise of the self-represented litigant who is using AI to formulate pleadings for unfair dismissals. I am not someone who will ever argue that you should make proceedings in any of our industrial relations commissions more legalistic. I think there's a lot to be said for our lay jurisdictions, like NCAT and the commission, where someone can go and represent yourself. There's a purpose to that. But it is genuinely clogging up our system that there are so many people who now tap out an answer in ChatGPT and then file that.

I was talking with one of my colleagues who works in this area and I said, "Well, why don't you just do what Screen Australia did and require declarations that 'My pleadings didn't use any AI' and perhaps run them through a reader?" That kind of information sharing doesn't seem to be happening across government. There is a consultative process, as I understand it, happening in the Department of Customer Service, which is Minister Jihad Dib's portfolio, if I'm not mistaken. But I suspect there is a way to have a whole-of-government information sharing, tripartite model where everyone's able to benefit from everyone's expertise, because AI is becoming such a generalist technology. There'll be things that Health are learning that I will never hear about unless there's some way of sharing across portfolios. That's the short list, but I think if we were able to sit down and write out a list of the harms it actually wouldn't be that difficult to reverse-engineer solutions for each of them.

The Hon. MARK BUTTIGIEG: Before I get into a broader question about the implications of the tripartite model in dealing with it, those large language models that are affecting your members in the sense that people can plug into the way language is used, even in a creative way, obviously then supplanting your members' jobs—can you just tease that out for us about how ubiquitous that's been?

CLAIRE PULLEN: It's a really huge problem and it's emerging in some of the research, particularly out of the US, about just how easy it is to reproduce copyrighted work. There's a particular team in the US that just released a study that has shown that, without mentioning the author name or any of the characters, just using semantic descriptions, you can get 90 per cent of a copyrighted work spat out verbatim from a large language model. That is then a competing product that is cheaper than buying the actual book or the actual screenplay or the score or the piece of music from the original creator.

In terms of my royalty collection work, it means that there's no creator in there being paid any royalty for that work. Royalties usually aren't huge amounts of money. There are some notable exceptions to that, but for every time someone watches an episode of TV, if you've been writing TV for 30 years, those little amounts add up over time. We do distribution once a year just before Christmas and the regular feedback from members is "This is perfectly timed because now I can go and buy the Christmas gifts." If there's a competing product, which will necessarily be cheaper because there are no royalties paid and there are no creators involved, people will choose that. Again, I don't think regulating consumer behaviour is actually the solution here. It is a rational economic decision to pick the cheaper option.

The Hon. MARK BUTTIGIEG: By way of example, if there's a successful series on Netflix—I can't remember some of them now; I know I've watched them—and there's the formula, so let's use one of these models to generate something similar which has got a known success, how do you unpick that? How do you work out that it's done that, and then how do you protect against it?

CLAIRE PULLEN: Look, we're reasonably confident that it's happened, and I'm happy to take you through how we're very confident that our members' works are in the large language models. One of the ways that I've tested this is you can actually go onto ChatGPT or Claude or one of the large language models and go, "Can you please write me the first scene of a Van Badham play that's about this character, that character and this one?", and you get one of Van's works. You can do this for Leah Purcell and you can do it for David Williamson. If you can name the Australian author or playwright, it can do it. It knows their voice well enough. Like I said, I'm happy to take you through the explanation of how we know the works are in there. One of the things that came out during the Federal inquiry into the safe and responsible use of AI in 2024—the big tech companies were using phrases like "publicly available", which is not a term at copyright law.

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What was revealed is they have programs that listen to things like YouTube or the ABC or SBS that are publicly available. That transcript then gets fed into the model to teach it the relationship between words, some of the scene breakdowns—that kind of work. "Publicly available" doesn't mean it's free and that someone doesn't own it. We all pay for the ABC, for example. YouTube has an entire business model structured around how advertisers are paid and the engagement with creators. There is no way for any creative—not just a writer, but a director, an editor, a composer or a set designer—to compete with a product that is essentially free, because you can have the most junior person in a production company going, "I would like to see a high school drama set in Australia, based on this existing property." There's no way that the entire team of creatives that Netflix spent a lot of money investing in, which paid off as an investment because it's one of the most watched television shows that Australia has ever produced—you just can't compete. There's no way to do it.

The Hon. MARK BUTTIGIEG: That leads to a broader question, and I've raised this in Parliament before and I know it's been the subject of an article in *The Australian Financial Review* today. This is a classic example of, I suppose, the coalface effect of what could happen right across the economy. That's a clear case of jobs being displaced as a result of not pre-empting what was obviously going to happen. What should be our institutional response to forestall that—not in a bad way that tries to restrict it, but bringing everyone on board so you get the benefits and the job security.

CLAIRE PULLEN: I think there are a couple of different ways to approach it. One is that there are clear frameworks for workers to bargain around what happens in their workplace, both at a Federal and a State level, for public service employees. Turning some attention to how those work to protect those people who are lucky enough to have those protections in place is useful. My members don't have those. We're not employees; we're contractors, for the most part. A large part of this is copyright and the Federal Government needing to do some work there, which is underway—it's not happening speedily, but it is underway—around addressing that original harm of the works being taken.

My members have quite a poor view of the initial theft. I think that informs their view that they don't want their works to be licensed. They're aware that is a potential option, but they're not super keen at the moment. I suspect some of that view might change if there was some way of dealing with the original theft—that we were dealing on even ground. I think that's a large part of it, but also, when I speak to my members, writing can be a profoundly isolating and internal process. It's just you and your laptop and a lot of feelings a lot of the time, and out of that we get beautiful plays and screenplays and audio dramas.

They talk to me about the isolation being worse when they're faced with a relentless propaganda campaign about the benefits of AI: It's going to make medicine safer, it's going to come up with all these new, exciting ways to exist in the world and the job replacement will all be fine, it's all going to come out in the wash. Some indication that they're being listened to by their governments I think would be incredibly useful to share the things that they've already learned. My members are not Luddites. Some of them are using AI. They're really clear about what they use it for and how they use it because writers tend to be early adopters of technology, so it's just about governments having those structures in place to balance the might of big tech, frankly.

The Hon. MARK BUTTIGIEG: I have got more but I'm conscious of not wanting to monopolise.

The Hon. JACQUI MUNRO: Thank you so much for coming. Yes, definitely, I think we have a strong alignment on a number of things. Definitely the Central Coast film studio. I don't think it has to be a one or the other situation. I think we can have both and I think we should. That's what good government is all about. I was curious about this idea of what you've spoken about in terms of data centres having responsibility for the data that's being held. I guess it's an idea like a tech supply chain assurance system, something like that. Can you tell me more about where that has operated overseas? I believe Norway has a pretty interesting system, but I'll let you—

CLAIRE PULLEN: I'm not familiar, except at a very surface level. Like with many things, the Scandinavians are doing this properly and well and taking a sovereign approach. Like I alluded to, if this is worth doing, we're going to get behind it in a fulsome way. It's not that unusual I would just say, just as a matter of principle, to require supply chain clarity in a lot of things, particularly around compliance with the law. Just because a house was built compliant with the planning laws at the time, it doesn't mean you won't address it when it later becomes a meth lab if that's the use that someone puts it to. I would say it's incumbent on governments to think about the entire life of a data centre. I think you're right—no doubt we could have both. I suspect the land that I would like to see a film studio on is not particularly as useful for a data centre as the other sites for it.

The Hon. JACQUI MUNRO: Yes. Exactly.

CLAIRE PULLEN: But I do have to ask the question: Why are we building a film studio to make very expensive films and TV here in Australia when—

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The Hon. JACQUI MUNRO: Also supported by an international company as well.

CLAIRE PULLEN: That's right. We could—and perhaps should—at some later date litigate the question of the tax incentives.

The Hon. JACQUI MUNRO: Indeed.

CLAIRE PULLEN: But leaving all that aside, down the road there's a data centre that is actively listening to films and videos lawfully and unlawfully and then creating a competing product where someone, again, sitting in office can go, "Can I please have a competing movie?"

The Hon. JACQUI MUNRO: Obviously, APRA AMCOS is doing work in their field of expertise as well. Is there a kind of broad creative industry copyright effort that is engaging with data centre operators or peak body—well, now there is a peak body—or even operators like it might be Anthropic or Google directly? You said that they haven't reached out to you but are you reaching out to them?

CLAIRE PULLEN: Yes. We have been and did after the Senate inquiry that I referred to, in 2024. There is a broader group. I'm actually missing our regular meeting to appear here today. I gave my apologies via APRA AMCOS like, "Nick, can you please let them know that I'll be late?" We are working very closely together, not just with creator groups, but with rights-holder groups. Our ecology is complex. In the way that journalists work with news media organisations, we work with production companies who then work with a motion picture association and the streamers, the broadcasters like the ABC, but also the commercial broadcasters. It has been a really heartening time of agreement from everyone about the idea that we do need to maintain the integrity of our intellectual property framework so that everyone can continue to benefit from that work.

Yes, we're working very closely together. We meet very regularly. APRA AMCOS are an incredible part of that work that we are doing, and we are regularly seeking contact with the big AI companies and the big tech companies. Unfortunately, I suspect that the litigation that's going on in the United States precludes them from reaching deals here in Australia in case it prejudices the proceedings overseas. That is just a suspicion, but that would be my view about what might be happening.

The Hon. JACQUI MUNRO: Do you know if that extends to data centre operators that are Australian, like NEXTDC or CDC?

CLAIRE PULLEN: I am not aware. As I suspect you know better than me, the idea of a peak body or one representative for the data centres is relatively new. I haven't initiated contact with them yet.

The CHAIR: The ICIP, which I know you've mentioned in the submission in passing, can you explain to us what that is?

CLAIRE PULLEN: Indigenous cultural and intellectual property. It is an area of copyright law that is changing. Copyright law is based on the idea of an individual owner of an individual property. That doesn't necessarily account for collectively owned or collectively transmitted works, like a lot of First Nations works, where a community or a moiety or a clan as a whole may own a story or a dreaming or a songline, rather than individuals owning it. There is some work happening at a Federal level around how we might incorporate this into the copyright system. It particularly comes up around the issue of fake Aboriginal art, which I'm sure everyone is aware has been a real issue in the last couple of years.

I hate to be able to say this, but you can actually use ChatGPT or Claude to create fake dreamings and songlines. I did this both using Australian indicators of First Nations storytelling and for Māori, to look after our New Zealand members, which indicates quite strongly that there are works in there that have been taken without permission and consent. I think about how hard, particularly, our broadcasters and our news organisations work to make sure that when you refer to Elders who have passed or First Nations people who have died, about how they're referred to by name—that their names are given correctly—about warnings before you show photos of Elders who have died and members of community.

AI has no guardrails around any of that. You can just type in someone's name before they died, get a photo of them and incorporate them into a story. You could have someone who is on the other side of the world going, "I'd like a minor Aboriginal Australian storyline in this TV drama. Can you just plot out some lines for me in the voice of someone who lives in this part of Australia and is from this particular community?" I don't see how the models, as they are currently operating, can be appropriately respectful of First Nations storytelling and collective ownership without data transparency and, frankly, some prevention of use in certain ways.

One of the things we talked about in other fora—you can use writers' names as a prompt. You can get on a large language model and go, "Can you please give me the next David Williamson play? Can I please have a Hans Zimmer score for this particular item?" You could say, just to choose a recent example, "The Presets

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covering Midnight Oil, where Peter Garrett joins at the end of the set." It's the same problem turned up to 11, I think, is the short answer. There are no protections in any of these models for First Nations traditions of storytelling that don't fit neatly within copyright frameworks, because the companies are ignoring those frameworks anyway.

The Hon. MARK BUTTIGIEG: It's interesting—you've got all this demand for unlimited desires. Whatever is happening in my head, I'll ask it and it can do it. It would be theoretically possible to turn that on its head, wouldn't it, and say to government, "Why don't we just write algorithms that say whenever someone's stealing IP, there's a flag that comes up and prevents you doing it"? You alluded to the legal thing before about "You're a citizen of New South Wales," but if I jump onto Copilot and say, "Write me a thesis on creative destruction and AI," and it's basically borrowing from someone else's paper, then theoretically there could be an algorithm to say, "You're actually stealing someone else's work, mate. You can't do it."

CLAIRE PULLEN: I think the issue we're all grappling with is that everyone's playing catch-up with the tech companies on this work. As part of the ACTU working group, we're working closely with the NTEU. Obviously how our work is cited and referred to is critical to an academic's career. Being correctly understood, referenced and prominent is really important to academics both in their work teaching and their work in developing research. I'm reluctant to suggest that we do anything that looks like regulating the consumer or the worker, for example, in the New South Wales public service, or working in New South Wales on the other end, and preventing what they can or can't type into something.

I think if you're regulating at that stage, it's probably too late, and the onus should be put on the companies to do it. Whether or not that's through an algorithm that detects and prevents, or they take the works out of the models, I can't say which of those would be easier or simpler, but I think the obligation should be placed on the company to do it—rather than prevent. There may be legitimate reasons, particularly around the educational use of works. There is a licence for government use of copyrighted works. There's news, there's commentary and there are parodies. There are exceptions at copyright law to using copyrighted works. Like I said, I don't expect normal people to get across the nuance of those details, but we could ask these giant international companies to do some work on it, I suspect.

The Hon. MARK BUTTIGIEG: The ACTU's position is quite strong on this, and they're quite conscious of it and publicly active. Do you think there's a sense that it's all too big and moving too fast, and we're just going to have to see what happens? It's sometimes the feeling I get.

CLAIRE PULLEN: Yes, I think that's right. When I talked earlier about AI being a general-purpose technology, and when people talk about how big it is, how it's going to change everything, how it's on everyone's computer, how it's on everyone's phone, the conclusion you're meant to draw is that it's too big to regulate and there's nothing that can be done. I just don't accept that that's the case. If you can articulate specific harms or likely harms—and I think that's why a tripartite approach is really good, because workers know their workplaces better than anyone else and can say to you, "This is the best way to look at what might happen out of this"—if you can articulate those harms, you just address the harms and then you are left with only the beneficial uses which, in an ideal world, is what happens with other general-purpose technologies, like cars, electricity and the internet. I don't accept the proposition that it's too big and therefore you shouldn't regulate. It's that second part that people don't say—the "Therefore you shouldn't regulate it". It may be big, but I have no doubt electricity felt the same.

The CHAIR: And it's powerful. If we take a step back then and we look at the extent to which data centres or the so-called data centre boom is perhaps not something that's critical infrastructure for us as people but certainly is critical infrastructure for AI as a technology and all of the companies that benefit from that, and if you were to redesign that, looking at what we have now and thinking about how you could do it better—if we accept that there's no stopping it—how do we get some of that benefit to come back to those who are actually the original labourers for that content?

CLAIRE PULLEN: I'll only talk about the sort of models that my members' works are in, because I'm certainly not going to stray into how it's being used in the medical field—not an area of expertise I have. Licensing is a really simple and straightforward system we already use. That's how we collect royalties and distribute those royalties to my members as income. There are movements, particularly in Europe, around the disgorgement of works from models and transparency regimes, so that creators can see what works are in models and what's not in there. There's a professor working out of South Australia who has been able to demonstrate that the use of AI in the creative industries is actually not going to be that prominent—he thinks. Or certainly that's what I've taken from his research. Because while it can meet an average person's creativity level—from my example, it's not that much better or worse than my writing, but it's way worse than one of my members, because they are professionally creative and that is their skill set.

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It's worth thinking about whether we actually need to use it for everything. There's that meme that is going around: "I really wanted the robots to do my washing so that I had time to write. Instead, the robots are doing the writing and now I'm doing all the chores." It's worth just being thoughtful, particularly at a government level. But every workplace is going to have to grapple with if this thing is actually going to do something that's beneficial. A lot of the research that's coming out is showing that the net outcome is actually lower productivity, particularly in the law, which is an area I pay attention to. If you replace your articulated clerks and your interns with AI, yes, you're not paying those people's wages and they're gone, but then the senior lawyers and the partners spend, on average, a certain percentage more time checking the work to make sure it's not making up cases or citing the wrong precedent or misunderstanding the legal issue, and so the net cost is actually higher. Particularly from government, I would like to see less starry-eyed buy-in around the propaganda, because there are people who believe the hype and think, "Just build it everywhere and see what happens."

The CHAIR: We are out of time. That was really very useful and incredibly interesting. To the extent there are questions taken on notice or supplementary questions, the secretariat will be in touch. That concludes our time with you. Thank you.

(The witness withdrew.)

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Miss ANNA JORDAN, Director, Protocol Policy Lab, affirmed and examined

Mr DEVJEET MATTA, Research Officer, Protocol Policy Lab, affirmed and examined

The CHAIR: Would you like to begin by making a short opening statement?

ANNA JORDAN: We would like to thank you, first of all, for inviting us to be here. Protocol is a youth-led technology and AI policy research lab, with the purpose to give young Australians the opportunity to join the national conversation on tech policy. Dev and I, along with a group of recent university graduates and university students, started Protocol because we saw a real necessity for youth-led and youth-informed tech policy. Technology, as I'm sure the panel is aware, has a huge impact on our lives. Its effects can be seen in many aspects such as education, work and daily life. That will continue to evolve.

Young people are among the most affected by tech policy decisions that are made today, which is why young Australian voices in tech policy matter. As young people with a long-term future in New South Wales, we care about long-term planning and sustainability. In our submission we compared approaches by jurisdictions in Australia and also overseas. In comparing their policies, we identified three key takeaways in their policy responses to data centre development. We think this inquiry presents a really significant opportunity for the New South Wales Government to implement a proactive policy response that prioritises sustainability without compromising innovation.

The CHAIR: I'll start with a few questions. My apologies: You are, as you said, a youth-led organisation, and I don't want to just ask questions of you as a young person; however, I'm going to start with just that. For some people over a certain age, you can be told that you're somehow anti-technology or that you just want things to be as they've always been. From your perspective as people who are completing university education and coming out the other side to see a world that is changing quickly because of AI, in terms of what is potentially a viable job, what is your view on AI and its displacement of jobs? How does that impact people who are just coming into the workforce?

ANNA JORDAN: Speaking from my experience and the experiences of the peers that I personally know, we are not anti-AI or anti-innovation but I think we see AI as an inevitable aspect of our lives. You're right: We're coming of age and entering the workforce and tertiary education at a time when AI is being adopted very quickly into these areas. Did you mention graduate—

The CHAIR: Yes. Where I'm coming from is, I speak with my own children, who had very clear ideas about the sorts of university degrees they wanted to do, but they are now saying to me, "I don't know if I can go into that field, because jobs will probably be all automated by then in that field, even creative fields." I'm just watching the speed at which they have to change where they think they might be employable, based on the speed of technological change, and I wonder if that's something you're seeing. Are people coming out of university with a degree now that they think is maybe not as useful as when they first went in?

ANNA JORDAN: I'm not sure I can fully answer that, because I am just speaking from my experience. I think there is for myself and my peers an acceptance that we need to understand AI and its impact on the industries that we've chosen. Dev and I are both law students/law graduates, so we've seen AI come into our lives in terms of more research-based applications. For us, it's been really important to be aware of those harms, but I personally haven't felt a sense of doom and gloom about my prospects. I can't really speak to the creative industries.

The CHAIR: One of the things I liked about your submission was the way you compared different jurisdictions; that's always incredibly useful. Can you tell us about your research, particularly of lessons learnt from other jurisdictions where they have had to jump in now to make fundamental changes to the way they do things? Can you talk a bit about that part of your submission?

DEVJEET MATTA: Just looking at other jurisdictions, we've found that a lot of other jurisdictions have projected that data centres are going to take a huge amount of additional energy from the grid, usually in five or 10 more years. What we've really seen is an assessment about which areas have that availability, in terms of renewable or hydrothermal energy—that kind of stuff. A lot of jurisdictions have made that assessment, and what they've tried to do is allocate areas where data centre providers can construct data centres. We've seen that in the UK, for instance.

There has been some policy where they've given incentives to data centres to build in areas which have access to, for example, wind energy. That's a big change that has been made. I would also say another large change is that some jurisdictions have also really had to assess the impact of job creation from data centres. I believe Victoria, for instance, has made a plan for that. The United Kingdom has sort of attempted to assess how major

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investments can help create jobs—jobs related to AI and tech, for instance. To sum up, there's been a real push for how renewable energy generation can help in constructing data centres and providing the often megawatts or gigawatts required to construct data centres.

The Hon. MARK BUTTIGIEG: On the job destruction piece, there are mixed signals. Intuitively, you get how there could be pretty economy-wide job destruction. But the actual study data coming back at this stage seems to be saying it's probably not as bad as we think at this stage. Do you have a view on that in terms of anything you've seen that might indicate the trajectory we're on? You hear these anecdotal stories that Atlassian sacked a thousand people or X companies got rid of so many admin jobs. There was actually an article in the Fin Review today that suggested that it's a bit more benign than people think at this stage, but it's one of those things where you would imagine the exponential trajectory could hit like a sledgehammer at some point. I just don't have a sense for how accurate the view is on that, in terms of whether there has been any modelling done on it. Is there a view on it?

ANNA JORDAN: Protocol hasn't done a comprehensive review, although I do know which conversation you're talking about. We would be very happy in the future to do research or collect data on that if the panel would be interested. Although I agree there is—I guess there are competing stories about layoffs or whether AI is really taking away jobs. We can't conclusively say.

DEVJEET MATTA: I agree. I think most of these figures tend to be projections. From the UK, for instance, I think it's 5,000-plus jobs are projected to be created in the near future. I think it's, in Ireland, 75,000 new jobs—and that's just from their strategies. But I think most of the data that we have does just tend to be estimations, and there's nothing concrete as to that.

The Hon. MARK BUTTIGIEG: Are you aware of any other jurisdictions where the job destruction has been more pronounced based on a greater uptake?

ANNA JORDAN: I couldn't say.

The Hon. SCOTT FARLOW: In terms of your report, your recommendations and your assessment in terms of other jurisdictions, what seems to be a key theme is that you need a strategy when it comes to data centre development. In terms of those key learnings you've got from all of those jurisdictions, what would you advocate as hallmarks of those strategies that New South Wales should be putting front and centre?

DEVJEET MATTA: From some of our recommendations, I think we suggested that there should be a strategy that balances, obviously, sustainability with speed. One thing I find interesting, looking at all the other jurisdictions, is some have allowed data centre providers to bypass local council regulations and planning laws and apply directly to either the State or Federal government for approval. So I think there's a real urgency in trying to get these data centres approved and sort of ride the AI wave. But at the same time I think it's important to keep in mind the sustainability metrics that are important. For example, Singapore, given the very constrained land size and given the obvious burdens on energy, has only approved data centre applications to providers who can meet certain quality benchmarks.

In terms of recommendations for New South Wales, I think providing approval for applications is important and ensuring that we do get a lot of data centre approvals in the next 10 or 20 years is definitely important. But I think that should be done while taking into account sustainability, making sure that sound acoustics is fine and doesn't impact the local community, ensuring that the centres can be appropriately cooled because obviously heat production is a huge issue, and ensuring we have the best IT equipment, the most sustainable IT equipment, is definitely important. I would say the biggest thing is striking a balance between those two things.

The Hon. SCOTT FARLOW: You used Singapore as an example and I think you outlined that it's got 54 per cent of the data centre load in Southeast Asia, so even with those standards in place it's not like people are saying, "Well, sorry, we're going to go to Malaysia instead," or, "We're going to go to Indonesia to operate." I take it that there are probably more international corporations that have their regional headquarter bases in Singapore than in surrounding areas, but are there other reasons for the viability or the interest in Singapore as a data centre hub?

DEVJEET MATTA: Yes. I think one of the reasons is, like you mentioned, it's a huge hub for AI and tech. I think another key thing is concentration. Singapore obviously has a small land size and that's not exactly comparable to New South Wales or Australia, which is quite large. But I think a real concern is clustering, and not just Singapore but other jurisdictions we've mentioned have obviously also granted approval in certain clusters of the city where they have access to the grid. So Singapore, along with other jurisdictions, potentially has that issue of clustering. I know in New South Wales we have a lot of data centres that are concentrated in certain areas of Sydney. I think that's an issue just generally to watch out for. Heat production is also a very big thing.

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Obviously, Singapore has a tropical climate and we get pretty hot summers here in Sydney and New South Wales, so that's another point of comparison.

The Hon. SCOTT FARLOW: In terms of that clustering, it's something that the industry wants and it's something that the industry sees advantage in, as you say, in terms of energy availability being there, but there are, of course, by-products and consequences of that. One of the areas in Sydney where there are a lot of data centres is Macquarie Park. Macquarie Park has had a metro line constructed to it. There are a lot of competing land uses there. Arguably, with such transport infrastructure around, data centres may not be the ultimate land use in that area. I think you mentioned the UK examples, particularly in terms of targeting centres. What were the characteristics of those centres that they targeted in the UK and is that something that can be applied in Australia potentially?

DEVJEET MATTA: Just give me a moment and I'll get my notes. The UK had similar, I think, metrics of approval to Singapore. We found that the approval was granted for areas they call AI growth zones. These are essentially areas in which there's priority access to grid connections. There are certain incentives for building there; there are fast-track planning approvals. So it was, in terms of location, areas where there would be improved wind generation capacity, areas that would help with the efficiency of the grid, and data centre providers who were happy to build in areas that would be more sustainable and more environmentally friendly.

The Hon. SCOTT FARLOW: Looking at those locations—Oxfordshire, Blythe, Cobalt Park, North Wales and South Wales—they're not typically the sort of population hubs of the United Kingdom as well. That potentially may be one of the challenges with data centres here, that they are competing in an area that we would want for housing or job opportunities, and of course noting the much smaller footprint of the UK compared to Australia, how would we be able to encourage, do you think, location of data centres perhaps in renewable energy zones, or perhaps closer to energy generation, rather than population centres?

DEVJEET MATTA: It's a difficult question. I wouldn't have all the answers to that. I think really there should be a focus on areas where renewable energy is sufficient. We have, obviously, in New South Wales particularly a lot of areas where we get a lot of sun; we get a lot of solar energy. I think at this stage we have something like 90 data centres in New South Wales. It's a bit hard to tell. Obviously most of them would be in Sydney. I think it's just a question for policymakers. What I can say is that I think in other jurisdictions they've been able to avoid the broader ramifications of building it in population-dense areas through things like sound controls, things like regulations around heating and cooling, and I think disincentivising providers who go outside of certain benchmarks. It's a difficult question, but I think that's all we can say based on that research.

The Hon. MARK BUTTIGIEG: Further to Scott's point, the market is going to take some of this, isn't it, in the absence of government intervention? Because there must be some sort of economic trade-off between—if you build them in renewable energy zones, for example, because you want cheap, affordable, clean power, is there a cost then of the latency, the speed with which you can transmit the data, because you're too far away from where the demand is? Then the closer you get to where the demand is, the more expensive the electricity becomes and the less renewables. Is there a view on that, is that that trade-off?

DEVJEET MATTA: Yes, I guess we would just reiterate that it is all about balance. I would point to the example of the US, where they haven't really been able to strike that balance as well. For instance, in our opinion, or through our research, they've let the policy happen on a very decentralised level, on the level of local states. What you've had is this issue of energy gentrification, which you also see in Ireland. As you mentioned, what you end up seeing is the price gets passed on to consumers and these data centres which are concentrated in these population-dense areas—energy becomes more expensive. That's one of the byproducts of providing very heavy tax incentives to data centre developers. Yes, I think it's really just about striking that balance. That comes out of our research, I'd say.

The Hon. MARK BUTTIGIEG: By contrast, the Nordic model—just tease that out for us a bit vis-a-vis America, where it's just decentralised and you end up clustering around population centres. What's happened in the Nordic countries?

DEVJEET MATTA: I think one of the benefits of Nordic countries is, I believe, in contrast to the US, they're all a hundred per cent renewable. They use 100 per cent renewables for their energy grid. You can correct me if I'm wrong on that. I believe their model is a lot more federal rather than the decentralised model used in the US. Just some examples, they have something called the Norwegian Energy Act, and they use that to set certain grid connection planning requirements. I think another really important thing is getting transparent information from these data centres. One of the benefits, obviously, of having a central regulation or policy to manage data centres is you can very easily get information from all these data centres on how much power they're using. So they have mandatory registration for all data centres which use an electrical capacity more than 0.5 megawatts. They have certain supervision powers. They have a disclosure requirement for how much

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cryptocurrency mining that certain data centres are doing. I would say, quite similar to New South Wales, they've also prescribed it as critical national digital infrastructure. I know in New South Wales we have SSDs—I think State significant developments—so there's that focus on achieving net zero, just like we have in New South Wales.

The Hon. JACQUI MUNRO: Thank you so much for the information and for coming today. It's really, really helpful. I was curious about the Norwegian model, in particular, and that data processing infrastructure that meets European standards. If you could elaborate on that a little, because what you were referring to just then with crypto—are there examples of what we were talking about in the session earlier around a kind of data supply chain, essentially, where you put in rules throughout the use of data, whether it's storage or collection or even analysis, and how that relates to the regulation of data centres in Europe?

DEVJEET MATTA: I think that's a little bit outside the scope of our research. Is it possible for us to maybe take that on notice?

The Hon. JACQUI MUNRO: Yes, for sure. That would be really helpful. I think Europe is an interesting example because you've obviously got individual countries that are responding to EU standards as a whole, and countries that have done that effectively. The EU is obviously known for its more stringent requirements, but if you've got countries that demonstrate that they're able to maximise investment, despite or potentially because of those requirements, that would be awesome to hear more about.

ANNA JORDAN: We'd be more than happy to give you that.

The Hon. JACQUI MUNRO: Just to be clear, it's helpful for the Committee and for the report to really understand your position around AI as an inevitability, and your view that it's really a management exercise rather than a kind of "stemming the tide" situation. Could you elaborate a little more on your views on that?

ANNA JORDAN: As Dev said before, just taking New South Wales as our context, there are already data centres in Sydney, so creating a proactive policy response would be addressing current development, not future development. In terms of—I know I used that word "inevitability" before, we are already in the stage of developing data centres so, yes, I think—

The Hon. JACQUI MUNRO: What are your concerns if we don't manage this effectively?

ANNA JORDAN: Keeping in mind Protocol's interest in policy development with data centres is to promote sustainability without compromising innovation, there's a concern that sustainability might not be protected or be prioritised with data centre development. Like Dev was saying before, it's all about balancing that priority of sustainability with interest in economic growth. Just speaking generally here, AI investment has happened very quickly. Just from observing the world around us, I don't think it's really a question of whether data centre investment would happen or not. Do you have anything to add?

DEVJEET MATTA: Yes, I'll pretty much just reiterate what Anna said. One of the major issues that we can foresee is the adequate use of space and making sure that there isn't that sort of clustering going on. I think that's one of the big issues. I would say we can all understand AI is inevitable, and the use of AI in schools, universities, workplaces—that's important. It's really key to appreciate balancing certain environmental standards—we think Norway has done a good job and, to a certain extent, Singapore—with job creation, with the appropriate use of AI. That's really the balance that we're looking for.

The Hon. JACQUI MUNRO: On that point about Norway and Singapore—and maybe more particularly Norway—doing a good job or a better job on that sustainability piece, do you think that is actually attractive to investors and data centre operators?

DEVJEET MATTA: I would say it's attractive to the extent that there needs to be a firm stance on which operators are given that energy to use. Certain jurisdictions have taken the approach where they're very lax in providing energy and approval for data centres to be created. Victoria is one of them.

The Hon. JACQUI MUNRO: As in Victoria's standards are lax?

DEVJEET MATTA: Yes, they are somewhat lax, from the research we have done. Norway and Singapore, both of them have set certain benchmarks. I think what's important is government takes a firm stance on which data centre providers are able to meet those benchmarks. If these standards aren't set and providing approval based on these standards, I think it is something that we will see—I think it is something that data centre providers will be happy to take on.

The Hon. JACQUI MUNRO: So it's a clarity issue?

DEVJEET MATTA: Clarity. Yes.

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ANNA JORDAN: I might add on. I'm not sure if the panel has seen the national expectations on data centre development, which is something that the Federal Government put out last month. Those, on the whole, articulate what standards the Government will be looking at when they're looking to approve future data centre approvals. I think that document similarly goes towards clarifying what the Government wants to see from future providers.

The Hon. JACQUI MUNRO: One of the criticisms when that came out was that it was potentially off-putting for some operators, but kind of also impotent in that there was no actual regulation, no penalty, for not complying with these standards. It goes to that lack of clarity piece. Is one data centre actually going to be approved over another if it doesn't comply or does comply with these standards? It was still fairly unclear as to what it would mean if a data centre operator didn't comply with those ideas.

ANNA JORDAN: It goes to what I was saying before with this being a good window for policy intervention in New South Wales and taking that proactive response.

The Hon. EMILY SUVAAL: You mentioned earlier, Mr Matta, in your remarks that the Victorian regulation was quite lacking, I can't remember your exact words. I'd just invite you to expand on that given the research that you've done.

DEVJEET MATTA: I think what I said was it was somewhat lax in terms of their approval. I will just refer to page 16:

CEO of CDC Data Centres, Greg Boorer, has described it as easier to get approvals for huge data centres in Victoria than elsewhere in Australia ... a NEXTDC facility in Port Melbourne was reportedly approved in 75 days, and a smaller facility in Geelong was approved in 62 days.

Whereas in New South Wales obviously the process for approval does seem to be quite slower. One was 19 months—a data centre by Goodman Group—while the next DC project approval took 26 months.

The Hon. EMILY SUVAAL: In terms of saying it's lax, I think what you're meaning is that the Victorian system is faster.

DEVJEET MATTA: Yes, that's what I meant to say.

The Hon. JACQUI MUNRO: Can I just clarify? So it's not necessarily that the restrictions are greater; it's just that the process is faster—sorry, the restrictions are less. Or the restrictions relating to environmental standards aren't necessarily lax or less stringent than New South Wales; it's just that the process is slower.

DEVJEET MATTA: Yes, I believe that's correct.

The CHAIR: I think it was in Singapore that they did a moratorium in 2019 and then followed that with actually having a strategy in place and then putting all of the guardrails on. In New South Wales I think a lot of people have gone, "Hang on a minute. We've got over 90 data centres already." It seems that there are no plans and structures; that has sort of just happened to us. Some submissions have recommended putting a moratorium on new data centre developments in New South Wales until we have that broader strategy in place. Do you support that?

DEVJEET MATTA: We have had some discussion about this, and that's not one of the recommendations we put forward. It's not to say that we don't necessarily recommend it, but we would suggest it's not as necessary for New South Wales as it is for Singapore. As a bit of context for Singapore, obviously their energy and sustainability concerns do tend to be greater than ours in terms of their spacing and heat. There are a lot of environmental issues regarding that. They have, I believe, nine more data centres than we do currently. But that's quite a lot given their small land size.

I think in Singapore's context the moratorium does make a bit more sense. I think for New South Wales it's feasible to develop a data centre plan alongside the construction of new data centres, so I'm not sure if that's something specifically we would suggest. Although, at the same time, the moratorium did allow Singapore to make what they call *The Green Data Centre Technology Roadmap* in which they would condition approval if they met certain parameters on sustainability and economic value and things like that. So, again, that's not something we would necessarily suggest, but it is something that they've pursued.

The Hon. MARK BUTTIGIEG: I just want to ask about—it's a related subject matter—the energy use and the competing claims. Obviously you've got this situation where, depending on who you talk to—the evidence we heard this morning is that it's not as dramatic as it's being made out. But let's say we're going to use a lot more electricity and water, whichever way you slice it, that means the appetite or the demand could get to the point where, if the market dictates that it wants a thousand data centres in two years or whatever, your electricity consumption goes through the roof. Renewables aren't enough to satisfy it at this point in time. Therefore it affects

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our net zero target. Have you got a view on those competing claims and whether or not those other jurisdictions are dealing with it better or worse, or where we are at? I'm just interested in if anyone's grappling with this.

DEVJEET MATTA: I think a good example of that might be the UK, which has come up with some regulations regarding that. Of course, at the same time, they have certain net zero commitments that they have to meet as well. I think one thing that they have done is they have organised local planning authorities. This is to designate certain locational requirements for data centres. If this is left to purely just the free market, obviously what would happen is you would have, as you said, potentially many more data centres being constructed at the cost of risks to the environment. Certain regulations where planning is taken into account—locational requirements are taken into account—for data centres is something that they've considered. Also considering them as essential infrastructure for flood risk purposes is something that they've done. I think I mentioned before AI growth zones as something that they've considered.

Certain other jurisdictions have also provided tax incentives but, again, this comes at a certain risk: like I mentioned, the problem of energy gentrification, where you have large clusters of data centres taking up large amounts of energy. That's some of the policies—the UK has done some important policies regarding that. I believe Ireland has also put in some regulations. I mentioned the tax incentives. I would also mention, from the data centre consultation plan of New South Wales, I believe one of the policy recommendations is that data centre developers fund their own infrastructure requirements. I think Ireland has actually done quite a lot on this. They have introduced a regulation which requires new data centres to install onsite dispatchable generation or battery storage. This essentially allows them to participate in the electricity market as a whole and requires them to take 80 per cent of their energy from renewable sources, so getting data centres to contribute to new renewable generation. That also helps to reduce some of the strain on the grid during times of high demand.

The CHAIR: That was excellent, thank you. To the extent there are questions taken on notice or supplementary questions, the Committee secretariat will be in touch. That concludes our time with you.

(The witnesses withdrew.)

The Committee adjourned at 14:30.